

Acute Stress Disorder Therapy by Virtual Reality: a Case Study of Ukrainian Refugees

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Figure 1: The VR-based therapeutic tool combines several therapeutic methods in a virtual environment, cutting the user off from reality. The tool was evaluated in a controlled experiment among office workers and during a pilot study among Ukrainian refugees.

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1 INTRODUCTION

According to WHO, stress is defined as any type of change that causes physical, emotional, or psychological strain [Organization 2021]. Stress is the human body's response to a disturbance that requires attention or action [Yaribeygi et al. 2017]. People experience positive and negative emotions that affect their interactions with others, perception of reality, and general well-being. How one responds to a stressful situation defines their life's comfort and health. Short-term stress mobilizes the body for effort and action, while long-term stress may cause harm. The symptoms of an acute stress reaction usually subside quickly, depending on a person's psychological ability to cope and recover. Untreated stress leads to more severe conditions, strains the nervous system, and causes somatic ailments [Bryant 2017]. That's why treatment and stress therapy is so important.

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Russia's armed aggression started on February 24, 2022. Neighboring countries were flooded with a wave of refugees dominated by women and children (the government in Kyiv has banned men of draft age from leaving Ukraine). Thus, according to various estimates, more than 2 million refugees from war-stricken Ukraine appeared in Poland in a short period. Women and children found themselves in an environment that was utterly unfamiliar, often without essential personal belongings, money, or a guaranteed place to stay. Separated from their loved ones, who remained in the bombed-out cities, these individuals were naturally highly stressed.

Having previous experience in using VR technology for stress reduction [Kamińska et al. 2020], we developed a Ukrainian-language VR-therapeutic application involving various relaxation techniques such as breathing methods, visualization, and bilateral stimulation (BLS). Furthermore, relatively minimally invasive biomedical sensors (dry frontal EEG, GSR, and induced eye movement tracers) enhanced the application to continuously monitor users' activities. Before and after the session, participants filled out questionnaires about their current mental health, stress levels, and general mood.

2 VR THERAPEUTIC TOOL

The proposed VR application is in the form of a cozy apartment in the mountains (see Fig. 1). During the session, participants relax on a comfortable chair (the same is mapped in the VR environment) in any convenient position (see Fig. 3a). The main scene is calibrated automatically (based on the user's head position) - the main element (a window with the mountain view) of the scene is centered vertically and horizontally. The narrator assists throughout the session acting as a therapist.

A short adaptation scene precedes the therapy scene. First, the user is asked to look around the apartment and get comfortable in

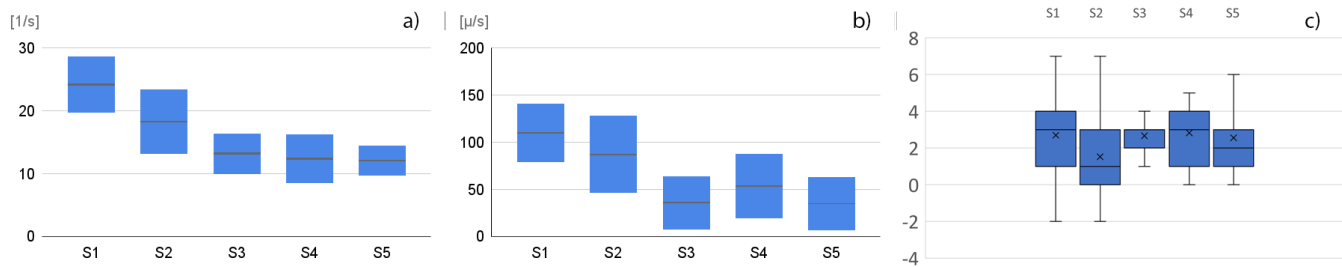


Figure 2: The box plot of a group analysis (average across all subjects for each session) a) of resistance gain over time b) of relaxation level gain over time (EEG) c) of loss of stress level - subjective feelings.

the chair. Then, the user moves seamlessly into the therapy sessions, which involve various relaxation techniques such as breathing methods, visualization, and bilateral stimulation.

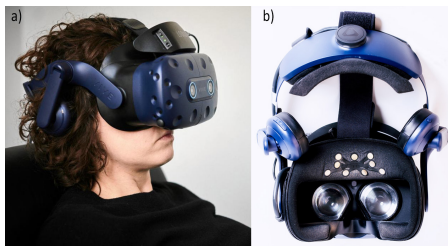


Figure 3: a) A snapshot of a subject during a VR therapeutic session. b) Looxid Link - VR compatible EEG sensor.

The sessions are monitored using EEG Looxid Link (see Fig. 3b) and GSR sensors. Those sensors allow the recording of raw measurement data and real-time monitoring of interpretation of the user's involvement and stress level. The system's unique feature is the capacity to compare data recorded during the user's subsequent therapy sessions. Thus, the therapist can easily monitor the progress and achieve effects. In addition, before and after the session, users fill out tests (GHQ12 [Goldberg and Williams 1988], GMS [Makowska and Merecz 2001]) so that we know their subjective feelings about their mental state at that moment, the intensity of stress and the emotions accompanying the moment.

Before each session, users were informed about the course of the VR experience and the exclusionary conditions, and provided informed consent. The entire course of the session was monitored by the supervisory staff. We assumed that each participant would be able to benefit from five therapeutic sessions (one session per week). Unfortunately, due to personal reasons and relocation, not everyone has completed the session sequence. Finally, we recruited $n = 55$ participants, 52 female, 3 male, aged $M = 35$, $SD = 10.93$. The study was conducted in a period when the university was closed (there were no students), in a quiet, air-conditioned room.

2.1 Results

To verify the effectiveness of the presented application, we used pre and post-tests (subjective measurement) and the GSR and EEG signal (objective measurement). The GSR sensor returns a vector of

resistance changes in time. First, we smoothed those vectors with a moving average (30 samples) to reduce measurement noise. Next, to determine the change in signal waveforms at the beginning and end of the session, the group analysis (average across all subjects for each session) in the form of mean and standard deviation values of resistance gain over time is presented in Fig. 2a.

Since the Looxid Link provides analytical information such as relaxation level (in a range of 0 to 1) over time, similarly as in the case of GSR, first we smoothed those information with a moving average (30 samples). Then, we determined the change at the beginning and end of the session (average across all subjects for each session) in the form of mean and standard deviation values of relaxation level gain over time. The results are presented in Fig. 2b.

The results obtained from pre and post-test, in the form of loss of stress level, are presented in Fig. 2c.

3 CONCLUSION

The drama of Russia's aggression against Ukraine and the massive refugee inflow prompted many Poles to help the displaced. As a scientific team with considerable experience in the field of stress reduction using immersive technologies, we decided to adapt our previous solution to those specific conditions. Several meetings with charitable organizations' representatives directly involved in helping refugees allowed us to develop a therapeutic scenario to reduce severe post-traumatic stress. The complete therapeutic sequence was successfully carried out on 30 participants within three months. The VR tool we produced is still being used by some Polish NGOs.

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